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Neuroprotective Potential and Chemical Profile of Alternatively Cultivated *Ganoderma lucidum* Basidiocarps

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Various neurodegenerative diseases are the main challenges to the modern medicine and there is a great need for novel, natural, neuroprotective agents. *Ganoderma lucidum* is a well-known medicinal mushroom, which health benefits have been confirmed by numerous studies. As demand for its basidiocarps is increased and traditional cultivation on hardwoods is not environmentally friendly and economically justified, finding of alternative substrates is necessary. The aim of the study was to assess the effect of alternative cultivation substrates on the chemical profile of *G. lucidum* basidiocarps and their capacity to inhibit acetylcholinesterase and tyrosinase, which higher activity is directly associated with neurodegenerative processes. Extracts of basidiocarps cultivated on alternative substrates, especially on clear wheat straw, showed significantly higher inhibition capacities than extracts of commercially-grown ones. These extracts were considerably different chemically from commercial basidiocarps extracts and even nine new compounds were isolated from them. Our results suggest that cultivation substrate greatly affect the chemical profile and neuroprotective capacity of obtained basidiocarps and wheat straw is a promising cultivation substrate.

Keywords: biological activity, chemical profile, cultivation substrates, enzymes, *Ganoderma lucidum*.

Introduction

The rapid development of medicine, as well as improvement of living conditions from the beginning of the 20th to the early 21st century, has resulted in increased life expectancy from 50 to 60 years to 80 to 90 years in developed countries.[1] On the other hand, the proportion of the population suffering from neurodegenerative diseases associated with ageing, such as Alzheimer's and Parkinson's diseases, is constantly increasing.[2] The etiology of Alzheimer's and Parkinson's diseases, the most serious neurodegenerative diseases, is not still completely understood, but it is assumed that oxidative stress has a significant impact on their early pathogenesis. The stress triggers a chain of events among which is increasing activities of acetylcholinesterase and tyrosinase that lead to a decrease of acetylcholine and dopamine levels that are directly associated with cognitive disfunction and loss of neurones.[2 – 4] Currently-available conventional drugs used to treat these diseases provide only a short-term delay of their progression and have numerous side effects, such as gastrointestinal disturbances.[2][5] Therefore, in the last few decades, there has been an increasing interest in finding new, natural sources of neuroprotective agents, which could be used as food supplements. Numerous studies have shown that some plants (such as *Narcissus* sp., *Mellisa officinalis*, *Vinca minor*, some vegetables, fruits and nuts) synthesise galantamine, an effective acetylcholinesterase inhibitor, preventing the occurrence of neuroinflammation and because of this effect such plants have been called a 'brain food'.[2] In contrast to our knowledge of beneficial plants, there are still scarce data on the antineurodegenerative potential of mushrooms.

The traditionally-known beneficial effects of *Ganoderma lucidum* usage, such as preserving vitality, promoting longevity, improving the nervous and immune system, eliminating fatigue, etc., have been confirmed by numerous modern studies and clinical trials. Thus, extracts and compounds derived from its fruiting bodies, spores, mycelia, and fermentation broth were shown to be powerful antioxidative, immunomodulating, antitumour, antimicrobial and antiaging agents.[4][6 – 11] Likewise, *Seow et al.*[12] reported that some species of the genus *Ganoderma* have the ability to enhance neuritogenesis and delay senescence

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of neurones and in such a way decrease the progression of neurodegenerative diseases. Because of their proven health benefits, *G. lucidum* basidiocarps have become a nutraceutical with increasing demand on the global health food market.

The fruiting bodies of *G. lucidum* have been commonly commercially cultivated on hardwood logs, stumps, and sawdust, although this is neither environmentally friendly nor economically justified. Therefore, its cultivation needs novel, more accessible, and cheaper substrates, which should also enhance biological efficiency of cultivation as well as bioactivity of obtained basidiocarps. As cultivation conditions are known to affect the qualitative/quantitative profile of active compounds in *G. lucidum* basidiocarps which determine their bioactivities,[13] optimization of the cultivation conditions, especially substrate composition, should have the crucial role. Wheat straw is one of the most abundant agricultural lignocellulosic residues in Europe, with only a small proportion currently used in biotechnological processes and the majority remaining as ballast in nature.[14] Previous studies have already shown that using wheat straw instead of existing commercial substrates stimulated antioxidative, antimicrobial, cytotoxic, and genoprotective potentials of various *G. lucidum* extracts.[7][15] Therefore, the aim of this study was to assess the effect of alternative substrates on the chemical composition and neuroprotective activity of *G. lucidum* basidiocarps.

Results and Discussion

Inhibition of Acetylcholinesterase and Tyrosinase Activity

Generally, all extracts of *G. lucidum* BEOFB 431 fruiting bodies showed considerable potential to inhibit acetylcholinesterase and tyrosinase activities compared with galantamine and α-kojic acid, respectively (*Figure 1,a, b*). However, the extracts of basidiocarps cultivated on the alternative substrates, especially those on wheat straw, showed significantly higher inhibition capacity compared with the extract of commercially-grown basidiocarps.

The degree of acetylcholinesterase inhibition by the extract of basidiocarps obtained on wheat straw gradually increased with extract concentration increasing from 0.05 to 0.8 mg/ml where an inhibition plateau (61.33%) was reached (*Figure 1,a*). The inhibition capacity of the extract of basidiocarps cultivated on the mixed substrate was also associated with increase in extract concentration, from 26.63% (at a concentration of 0.05 mg/ml) to 52.90% (at a concentration of 1.6 mg/ml). A significantly lower activity ($P < 0.01$) was observed for the extract of commercial basidiocarps which gave an inhibition rate ranging between 24.5% (at a concentration of 0.05 mg/ml) and 35.09% (at a concentration of 1.6 mg/ml). The effectiveness of extracts was also compared using IC_{50} values, which ranged from 0.56 mg/ml for basidiocarps obtained on wheat straw, through 1.42 mg/ml for basidiocarps cultivated on the mixed substrate, to 2.3 mg/ml for commercial basidiocarps. The synthetic inhibitor, galantamine, was more effective, especially at lower concentrations, which resulted in a significantly lower IC_{50} value (0.03 mg/ml). However, at higher concentrations (0.8 and 1.6 mg/ml), its inhibition capacity was slightly higher than the capacity of the extract of basidiocarps from wheat straw (67.15% and 70.06%, respectively) (*Figure 1,a*).

Regarding tyrosinase inhibition, the extract of fruiting bodies cultivated on wheat straw was also the most effective, i.e., at 1.6 mg/ml even 86.42% of the enzyme was inhibited. The extract of basidiocarps grown on the mixed substrate caused a slightly lower inhibition, ranging from 35.19% to 82.46%, while the extract of commercial basidiocarps showed the weakest capacity (from 25.7% to 69.35%) (*Figure 1,b*). Thus, extracts of basidiocarps obtained on the alternative substrates showed significantly higher inhibitions at lower concentrations, which were even greater than the capacity of α-kojic acid (33.93%), which was also reflected in their IC_{50} values. α-Kojic acid was a more effective tyrosinase inhibitor (IC_{50} = 0.18 mg/ml) than the extract of commercial basidiocarps (IC_{50} = 0.52 mg/ml), but weaker than extracts of fruiting bodies cultivated on wheat straw (IC_{50} = 0.12 mg/ml) and the mixed substrate (IC_{50} = 0.14 mg/ml).

This study is a rare example of establishing the capacity of a mushroom to participate in neuroprotection, thereby leading to possible treatments of neurodegenerative disorders. Likewise, to our knowledge this was the first study of the influence of substrate type and composition on the chemical profile and bioactivities of basidiocarps. Our previous study has already shown that various *G. lucidum* strains, cultivated on the same substrate, differed in numerous bioactivities, which was the consequence of their geographical and genetic diversity.[7] However, in this study we demonstrated that basidiocarps of the most active strain, cultivated on different substrates, were significantly different in neuroprotective capacity, which confirmed considerable influence of substrate composition on the bioactivity. *Hasnat et al.*[16] reported that a water extract of *G. lucidum* basidiocarps grown on germinated brown rice could inhibit acetylcholinesterase. Namely, the extract was a weak

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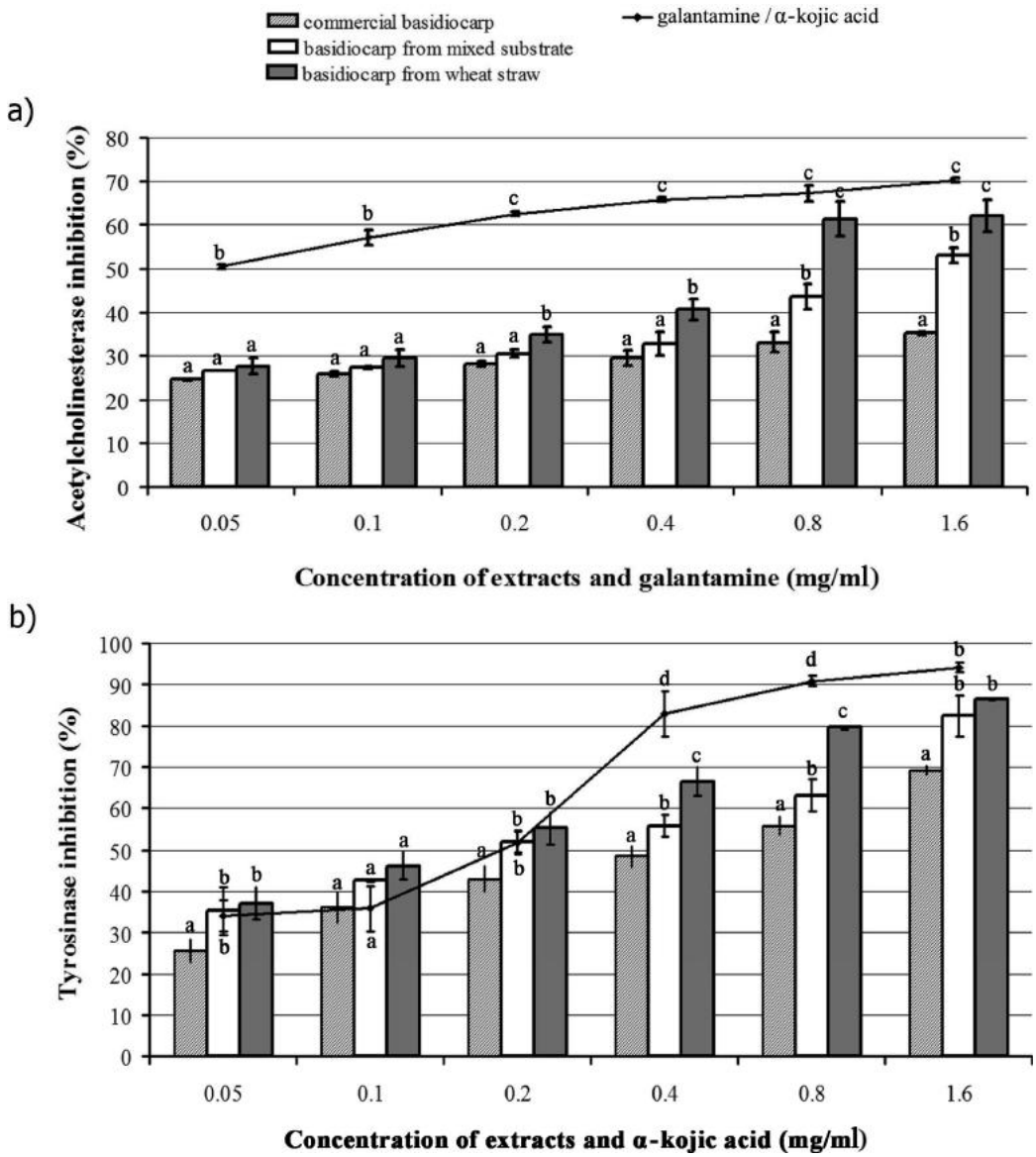

Figure 1. Capacities of extracts of *Ganoderma lucidum* basidiocarps cultivated on different substrates and synthetic preparations to inhibit acetylcholinesterase *a*) and tyrosinase *b*). Data represent mean value ± standard error, $n = 3$. The values with different superscripts (a,b,c,d) differ significantly ($P < 0.05$) according to Tukey's test.

enzyme inhibitor as it caused inhibition of 19.46% at a concentration of 0.125 mg/ml, and at even 2.0 mg/ml only 57.01%, which was reflected in its IC_{50} of 1.01 mg/ml. A weak inhibitory effect was also found for a *G. lucidum* basidiocarp water extract fermented by lactic acid bacteria, i.e., at 1.25 mg/ml about 33% of acetylcholinesterase was inhibited and at 5.0 mg/ml about 54%.[1] In comparison with our results, these inhibitory capacities were around only half those we found for the extract of basidiocarps cultivated on untreated wheat straw.

As well as high acetylcholinesterase inhibition, our extracts of *G. lucidum* BEOFB 431 were more effective in tyrosinase inhibition in comparison with various extracts of not only *G. lucidum*, but also other mushroom species. Fruiting body extracts of the *G. lucidum* strain used in the study of *Chien et al.*[17] showed slightly lower enzyme inhibition than the extract of basidiocarps cultivated on wheat straw, i.e., it inhibited 80% of tyrosinase at a concentration of 1.0 mg/ml. However, various extracts of other mushrooms were significantly weaker tyrosinase inhibitors. Thus, n-hexane, ethyl acetate and butanol extracts of *Pleurotus citrinopileatus* basidiocarps at 100.0 μg/ml inhibited the activity by 41.0%, 27.4% and 28.8%, respectively, while inhibitory action of acetone, methanol and hot water extracts of *P. nebrodensis*, *P. ferulae* and *P. ostreatus* ranged between 2.57% and 59.56%

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depending on the concentration.[18 – 21] The effectiveness of the extract of *G. lucidum* BEOFB 431 basidiocarps, especially those obtained on wheat straw, can be particularly demonstrated by comparison with the inhibition potential of *Inonotus mikadoi*, one of the weakest species to inhibit tyrosinase, whose extract at 10.0 mg/ml inhibited only 46% of the enzyme.[22]

From these comparisons, we can conclude that the strain used in our study, combined with the substrates we used, has significant potential to produce acetylcholinesterase and tyrosinase inhibitors.

Chemical Composition of Basidiocarp Extracts

Chemical analysis of the basidiocarp extracts showed significant differences among commercially-grown basidiocarps and those cultivated on alternative substrates (*Figure 2*; *Table 1*). Namely, only one compound was detected in all three extracts, while extracts of basidiocarps cultivated on alternative substrates and those obtained on a commercial substrate showed 13 compounds to be unique to extracts from the alternative substrates and another 13 unique to the commercial substrate. Although ganoderic, ganoderenic, and lucidenic acids were predominant triterpenoids in these extracts, they were not equally represented in the extracts (*Table 1*).

Compared with the available literature, only five compounds isolated from extracts of alternatively-cultivated basidiocarps have previously been identified (9,12,13-trihydroxy-10*E*-octadecenoic acid, Ganoderiol D, Ganoderiol B, Ganodermanontriol,

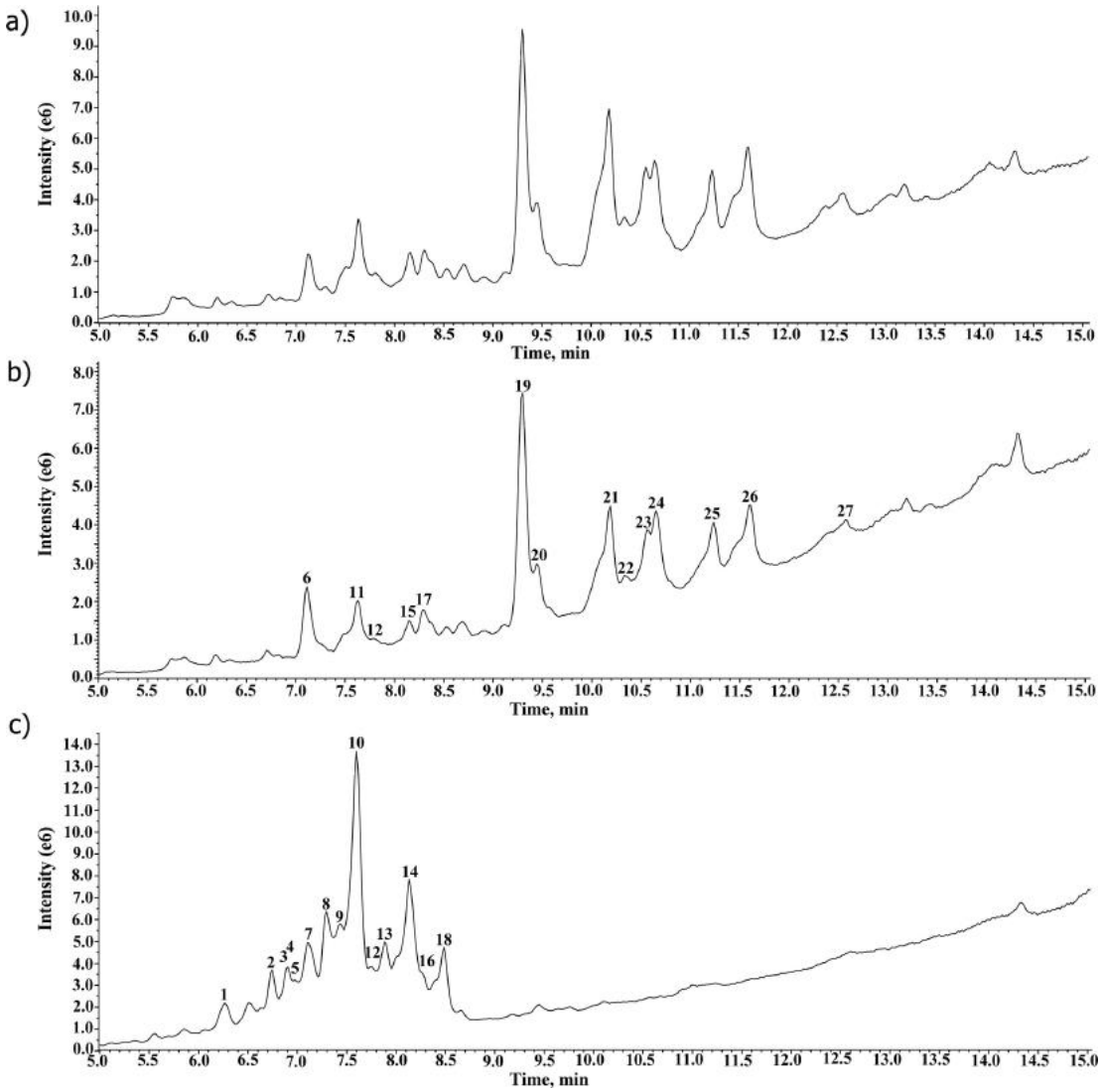

Figure 2. LC/TOF-MS chromatogram of extracts of *Ganoderma lucidum* basidiocarps cultivated on wheat straw *a*), mixed substrate *b*), and commercial substrate *c*).

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Table 1. LC/MS data of extracts of *Ganoderma lucidum* basidiocarps cultivated on different substrates

No.	Retention time [min]	Bruto molecular formulae/compound[a]	$[M-H]^-$ [*m/z*]	Adducts [*m/z*]	Commercial substrate	Mixed substrate	Wheat straw
1	6.23	$C_{30}H_{44}O_8$	531.2933	645.2901	+	–	–
2	6.75	$C_{30}H_{46}O_7$/Ganoderic acid C2	517.3228	563.3168; 1035.6395	+	–	–
3	6.88	$C_{30}H_{42}O_8$/Ganoderic acid C6	529.2790	1059.5669	+	–	–
4	6.89	$C_{30}H_{40}O_7$	511.2685	625.2548; 1137.5466	+	–	–
5	6.97	$C_{27}H_{40}O_6$/Lucidenic acid N	459.2737	505.2794; 919.5578	+	–	–
6	7.12	$C_{18}H_{34}O_5$/9,12,13-Trihydroxy-10*E*-octadecenoic acid	329.2324	–	–	+	+
7	7.15	$C_{30}H_{42}O_7$/Ganoderenic acid B	513.2840	627.2709; 1027.5785	+	–	–
8	7.28	$C_{30}H_{42}O_6$/Ganoderic acid GS-1	497.2892	–	+	–	–
9	7.39	$C_{30}H_{42}O_7$/Ganoderic acid AM_1	513.2848	559.2906; 1027.5780	+	–	–
10	7.59	$C_{30}H_{44}O_7$/Ganoderic acid B	515.2998	561.3051; 629.2987; 1031.6068	+	–	–
11	7.62	$C_{32}H_{42}O_9$	569.2741	615.2819; 683.2704; 1139.5586	–	+	+
12	7.85	$C_{30}H_{40}O_7$	511.2711	625.2581; 1023.5467	+	+	+
13	7.89	$C_{27}H_{38}O_6$/Lucidenic acid A	457.2578	915.5246	+	–	–
14	8.12	$C_{30}H_{40}O_6$	495.2742	–	+	–	–
15	8.16	$C_{32}H_{42}O_9$	569.2741	615.2819; 683.2704; 1139.5586	–	+	+
16	8.26	$C_{30}H_{40}O_7$/Ganoderenic acid D	511.2686	625.2631; 1023.5448	+	–	–
17	8.29	$C_{31}H_{52}O_8$	551.3579	–	–	+	+
18	8.47	$C_{32}H_{42}O_9$/Ganoderic acid F	569.2744	683.2712; 1139.5570	+	–	–
19	9.30	$C_{30}H_{48}O_5$/Ganoderiol D	487.3404	533.3466; 601.3366; 975.6912; 1021.6928	–	+	+
20	9.46	$C_{31}H_{52}O_7$	535.3600	–	–	+	+
21	10.17	$C_{21}H_{28}O_4$	343.1903	–	–	+	+
22	10.33	$C_{21}H_{26}O_5$	357.1703	471.1638; 715.3489	–	+	+
23	10.53	$C_{17}H_{24}O_2$	259.1679	373.1650	–	+	+
24	10.66	$C_{30}H_{46}O_4$/Ganoderiol B	469.3321	515.3365	–	+	+
25	11.23	$C_{30}H_{46}O_5$	485.3259	–	–	+	+
26	11.56	$C_{30}H_{48}O_4$/Ganodermanontriol	471.3493	–	–	+	+
27	12.54	$C_{30}H_{44}O_4$/Ganoderic acid DM	467.3159	513.3223; 935.6392	–	+	+

[a] Identification based on previously published data.

Ganoderic acid DM), while a further nine remain uncharacterized.

Despite identical qualitative chemical profiles of extracts of basidiocarps obtained on alternative substrates, their bioactivities significantly varied, which is likely to be the consequence of different proportions of the compounds.

Numerous previous studies have reported more than 140 triterpenoids, such as ganoderic, lucidenic, ganoderenic, and applanoxidic acids, which are synthesized by *G. lucidum sensu lato* and are responsible for various medicinal effects.[2][13] Considering that these compounds have been shown to have high antioxidative potentials and that oxidative stress is associated with the etiology of neurodegenerative diseases, their intake could prevent the development of the disorders. Further, the inclusion of natural neuroprotective agents in therapies would also have other benefits. Namely, conventional drugs that are commonly used in the treatment of these neurodegenerative disorders are effective only in the early disease stages and also lead to the organism developing tolerance to their long-term usage. This is not the case with natural neuroprotective agents.[23] Likewise, the characteristics of compounds found in *G. lucidum* basidiocarp extracts meet the main prerequisites for effectiveness of an antineurodegenerative agent, i.e., molecular weight, charge, and lipophilicity, which enable them to diffuse from the blood into the central nervous system and therefore overcome the blood-brain barrier.[23] As compounds identified in extracts tested in this study have molecular masses lower than 900 Da, thus classifying them as small molecules, and as they are lipophilic, they could be promising candidates for new natural neuroprotective drugs.

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Conclusions

There is no literature data on the influence of the cultivation substrate composition on the chemical profile and bioactivity of *Ganoderma lucidum* fruiting bodies. The results of our research clearly show that the composition of the substrate significantly profiled the chemical composition of the obtained fruiting bodies and thus significantly influenced their antineurodegenerative activity.

The substrate consisted of pure, unenriched wheat straw proved to be the most optimal on which the fruiting bodies of significantly different chemical composition and the largest neuroprotective potential were obtained, compared with the mixed and commercial substrates. Wheat straw has even induced the synthesis of new, in the available literature still unknown compounds of *G. lucidum*. The neuroprotective capacity of the obtained fruiting bodies was remarkable, even similar to commercial, synthetic drugs (galantamine and α-kojic acid). This is of great importance and the most important contributions of our work because wheat straw is one of the most available and cheapest waste, which makes it a promising alternative to the commonly used commercial substrates. These results imply the importance of further research in the area of the influence of the substrate on the bioactivity of *G. lucidum* basidiocarps in order to optimize it. Further studies will also go towards the isolation of specific bioactive compounds of *G. lucidum* obtained from wheat straw as well as their more detailed *in vitro* and *in vivo* studies. That should be a basis for the development of new pharmaceuticals based on alternatively and more cheaply cultivated *G. lucidum* basidiocarps.

Experimental Section

Organism and Growth Conditions

The culture of *Ganoderma lucidum* BEOFB 431 was isolated from fruiting body collected from Bojčin forest near Belgrade (Serbia) and maintained on malt agar medium in the culture collection of the Institute of Botany, Faculty of Biology, University of Belgrade (BEOFB). The collected fruiting body was determined as *Ganoderma lucidum* based on morphological features (pileus and stipe shape and colour and spores dimensions). Genetic identification was also performed on ITS, tef1-a and rpb2 gene sequences which were deposited in NCBI GenBank under accession numbers KX371594, KX371597 and KX371600. Fruiting bodies were cultivated under laboratory conditions by method of *Ćilerdžić et al.*[24] on two alternative substrates (wheat straw, and wheat straw/grape vine sawdust/wheat bran in ratio 8:1:1). The wheat straw and grape vine sawdust were obtained from the local agricultural holding. Basidiocarps of a commercially-cultivated strain as well as wheat bran were purchased at a health food store.

Preparation of the Fungal Extracts

Extracts were prepared in two steps: *i*) extraction of 10.0 g of dried and pulverised *G. lucidum* basidiocarps with ethanol (300.0 ml, 96%) on a magnetic stirrer (150 rpm) for 72 h and *ii*) centrifugation (20 °C, 1000 *g*, 10 min). The obtained supernatants were filtered through Whatman No. 4 filter paper, and the filtrates concentrated under reduced pressure in a rotary evaporator (Büchi, Rotavapor R-210, Switzerland) at 40 °C to dryness. The dried extracts were redissolved either in 5% dimethyl sulfoxide (DMSO) in sodium phosphate buffer (0.1 mM, pH 7.0) to an initial concentration of 1.6 mg/ml (for analysis of neuroprotective effect) or in methanol (HPLC purity) to a concentration of 10.0 mg/ml (for analysis of chemical composition).

Determination of Acetylcholinesterase Activity

Acetylcholinesterase (AChE) activity was determined spectrophotometrically using 96-well microtiter plates by the method of *Ellman et al.*[25] The reaction mixture [(0.1 M sodium phosphate buffer (pH 7.0), 0.01 M 5,5′-dithiobis(2-nitrobenzoic acid) (DTNB), extract, and AChE (5.0 U/ml)] was incubated at 25 °C for 15 min. The hydrolysis of DTNB was initiated by acetylthiocholine iodide, and the resultant production of yellow 5-thio-2-nitrobenzoate anion (TNB^{2-}) was quantified from its absorbance at 412 nm. The mixture without AChE was used as a blank. The degree of AChE activity inhibition was determined according to the equation:

$$\text{AChE activity inhibition rate } (\%) = [(E - S)/E] \times 100$$

E – enzyme activity without extract, S – enzyme activity with extract.

IC_{50} was defined as the concentration of extract inhibiting the enzyme by 50%. The obtained values were compared with galantamine, the commercial AChE inhibitor. All the chemicals were purchased from Sigma–Aldrich (Germany).

Determination of Tyrosinase Activity

Tyrosinase activity was determined spectrophotometrically using 96-well microtiter plates by the method of

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Likhitwitayawuid and *Sritularak*.[26] The reaction mixture, composed of phosphate buffer (pH 6.8), extract, and tyrosinase (46.0 U/l), was incubated at 23 °C for 10 min. The reaction was then initiated with L-DOPA, and the absorbance change was measured at 475 nm after 30 min of incubation. The rate of tyrosinase inhibition was determined according to the equation:

$$\text{Tyrosinase activity inhibition rate } (\%) = \{[(A-B)-(C-D)]/(A-B)\} \times 100$$

A – absorbance of tyrosinase in phosphate buffer, B – absorbance of phosphate buffer, C – absorbance of the reaction mixture, D – absorbance of the extract in phosphate buffer.

The extract effectiveness was defined by determining the IC_{50} value and comparing this with the value obtained for α-kojic acid, the commercial tyrosinase inhibitor.

All the chemicals were purchased from Sigma–Aldrich (Germany).

Analysis of Extract Chemical Composition

HPLC-DAD/ESI-TOF-MS analyses were carried out on an Agilent 1200 Series HPLC system (Agilent Technologies, Waldbronn, Germany) equipped with a degasser, a binary pump, an autosampler, a thermostated column compartment and a diode array detector (DAD) coupled with a 6210 Time-of-Flight LC/MS system (Agilent Technologies, Santa Clara, California, USA) via an electrospray ionization (ESI) interface. The chromatographic separation was achieved on a Zorbax EclipsePlus C_{18} column (100 × 2.1 mm i.d.; 1.8 μm). The mobile phase consisted of water containing 0.2% formic acid (A) and acetonitrile (B). A combination of isocratic and linear gradient modes of elution was applied as follows: 0 – 2 min 20% B, 2 – 30 min 20 – 95% B, 30 – 35 min 95% B, 35 – 36 min 95 – 20% B, and 36 – 40 min 20% B. The mobile phase flow-rate was 0.40 ml/min, the column temperature was set at 40 °C, and the injection volume was 2.0 μl. The spectral data were accumulated in the range of 190 – 450 nm, and representative chromatograms were recorded at 254 nm. The HPLC effluent was directed into the atmospheric pressure ESI ion source of the mass spectrometer. The eluted compounds were mixed with nitrogen in the heated nebulizer interface and negatively charged ions were obtained by applying the following ES parameters: capillary voltage, 4000 V; gas temperature, 350 °C; drying gas (N_2) flow, 12 l/min; nebulizer pressure, 45 psig (310.26 Pa); fragmentor voltage 140 V, and masses were measured in the range of 100 – 1500 *m/z*. A personal computer system running MassHunter Workstation software was used for data acquisition and processing.[27] The Molecular Feature Extractor of MassHunter Workstation was used to predict chemical formulas. Compounds were characterised by their retention times (t_R), mass spectra and UV spectra, and were identified based on previously published data.[28 – 30]

Statistical Analysis

The assays were carried out in five replicates and the results are expressed as the mean ± standard error. One-way analysis of variance (ANOVA) and Tukey's HSD posthoc test were performed to analyse any significant differences among means. Statistical significance was declared at $P < 0.01$. All statistical analyses were done using software STATISTICA, version 6.0 (StatSoft, Inc., Tulsa, USA).

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Author Contribution Statement

J. Ć., *M. S.*, and *J. V.* designed the study. *J. Ć.* cultivated *Ganoderma lucidum* basidiocarps on different substrates, prepared extracts and performed the tests for tyrosinase and acetylcholinesterase inhibition. *I. S.*, *V. T.*, and *I. B.* performed chemical analysis of tested extracts. *J. Ć.*, and *M. S.* performed statistical analysis and wrote the manuscript. *S. D. L.*, and *J. V.* contributed to the correction of the final manuscript version.

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